

5	(a) (i) (induced) e.m.f. proportional to rate of change of (magnetic) flux (linkage) / rate of flux cutting	M1 A1	[2]
	(ii) 1. moving magnet causes change of flux linkage 2. speed of magnet varies so varying rate of change of flux 3. magnet changes direction of motion (so current changes direction)	B1 B1 B1	[1] [1] [1]
	(b) period = 0.75 s frequency = 1.33 Hz	C1 A1	[2]
	(c) graph: smooth correctly shaped curve with peak at f_0 A never zero	M1 A1	[2]
	(d) (i) resonance	B1	[1]
	(ii) e.g. quartz crystal for timing / production of ultrasound	A1	[1]
6	(a) (i) $2\pi f = 380$	C1	